

Hoang-Khoi Do

3D GENERATION RESEARCHER · COMPUTER VISION ENGINEER

Dublin, Ireland

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Summary

I am a PhD student at the School of Computer Science and Statistics, Trinity College Dublin, Ireland, sponsored by an SFI research grant titled "Language3D: Creating Editable 3D Content from Deep Language Understanding for 3D-First Digital Platforms", led by Dr. Binh-Son Hua. My goal is to make 3D content generation more accessible by developing innovative, low-demanding, techniques that simplify the process for users and expand its applications. I was a research engineer at VNG, a unicorn in Vietnam. I finished my undergraduation with researchs in Image Segmentation and Long-tailed Learning.

Field of Interest

MY RESEARCH INTERESTS INCLUDE, BUT ARE NOT CONFINED TO:

- Conditional 3D Generation
- 3D Scene Reconstruction
- Surface Reconstruction
- Image Segmentation
- Long-tailed Learning
- Multi-task Learning

Education

Hanoi University of Science and Technology (HUST)

B.S. IN TELECOMMUNICATION ENGINEERING

Hanoi, Vietnam

Sep. 2020 - Sep. 2024

Skills

Deep Learning [Pytorch, TensorFlow, Keras]
Programming [CUDA, Python, C++, C, CMake]
Dev [Kubernetes, Docker, Git]

Work Experience

CV Research Engineer

GOTIT - SCANIT VIETNAM

Hanoi, Vietnam

June. 2023 - June. 2024

- Developed a **3D Geometry Restoration** pipeline for **document dewarping**, utilizing **Structural and Textual Line Matching** to correct perspective distortion.
- Training **light-weight** (< 5M params) models for **document information extraction, OCR, and classification** (~ 2000 documents/seconds).

CV Research Engineer

TRUEID - VNG CORPORATION

Hanoi, Vietnam

Nov. 2022 - Nov. 2023

- Researching on **document dewarping** by modeling the problem as a 3D surface reconstruction task, utilizing **Control Points Prediction** to correct geometric distortions.
- Secured a **\$1M R&D contract** with **Asia Commercial Bank** by designing a **lightweight**, high-throughput anomaly detection system, outperforming competitor benchmarks in accuracy and latency.

3D Software Engineer

ADT CREATIVE - ADT GROUP

Hanoi, Vietnam

July. 2019 - Sep. 2020

- Developing AR/VR/MR motion-tracking applications and games with **Unity** and **Vuforia**.

Publications

INTERNATIONAL CONFERENCE

1. **Khoi Do**, Binh-Son Hua, "Text-to-3D Generation using Jensen-Shannon Score Distillation," in *International Conference on 3D Vision, 3DV*, 2025. Accepted. To appear.
2. **Khoi Do**, Duong Nguyen, Nam-Khanh Le, Quoc-Viet Pham, Binh-Son Hua, and Won-Joo Hwang, "Domain Generalization via Pareto Optimal Gradient Matching," in *European Conference on Artificial Intelligence, ECAI*, 2025.

INTERNATIONAL JOURNAL

1. **Khoi Do**, Nguyen, D., Tran, N. H., and Nguyen, V. D., "PAT: Pixel-wise Adaptive Training for Long-tailed Segmentation," *Pattern Recognition Letters*, vol. 192, pp. 43–50, 2026.
2. Nguyen, Minh-Duong, **Khoi Do**, Le, Nam-Khanh, Tran-Nguyen H., Yang-Zhaohui, Nguyen Van-Duc, Chien Trinh-Van, "Toward Layer-Wise Personalized Federated Learning: Adaptive Layer Disentanglement via Conflicting Gradients," *IEEE Transactions on Networking*, vol. 34, pp. 3210–3225, 2026.
3. **Khoi Do**, Nguyen, D., Nguyen, H., Tran-Thanh, L., Tran, N.-H., and Pham, Q.-V., "Revisiting LARS for Large Batch Training Generalization of Neural Networks," *IEEE Transactions on Artificial Intelligence*, vol. 6, no. 5, pp. 1321–1333, 2025.

Teaching Experience

[CSU44052] Computer Graphics (Lecturer: Prof. Binh-Son Hua)

TEACHING ASSISTANT

*Computer Science and Statistics,
Trinity College Dublin
Winter, 2025*

[CSU23021] Microprocessor System (Lecturer: Prof. Quoc-Viet Pham)

TEACHING ASSISTANT

*Computer Science and Statistics,
Trinity College Dublin
Spring, 2025*

[CSU44052] Computer Graphics (Lecturer: Prof. Binh-Son Hua)

TEACHING ASSISTANT

*Computer Science and Statistics,
Trinity College Dublin
Winter, 2024*

[ET4248E] Machine Learning for Medical System (Lecturer: Dr. Viet-Dung Nguyen)

TEACHING ASSISTANT

*Electrical & Electronics Engineering,
HUST
Spring, 2024*

[ET2072E] Information Theory (Lecturer: Prof. Tien-Hoa Nguyen)

TEACHING ASSISTANT

*Electrical & Electronics Engineering,
HUST
Spring, 2023*

Certificates

- 2022 **Applied Generative Adversarial Network - Specialization**, DeepLearning.AI
- 2022 **DataBricks - Advanced Data Engineering with DataBricks - Specialization**, DataBricks
- 2022 **Advanced Computer Vision with TensorFlow**, DeepLearning.AI
- 2022 **TensorFlow: Advanced Techniques**, DeepLearning.AI
- 2021 **Google TensorFlow Developer**, Google TensorFlow

Academic Services

- 2026 **Reviewer**, NIPS2026, CVPR2026, 3DV2027
- 2025 **Reviewer**, ICLR2026
- 2024 **Reviewer**, ICLR2025